# C4.2 Transfers of energy and matter

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c4-2.html*

*Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 578–601 of the course textbook. Image credits are on the last slide of the slides for this subtopic.*

## ENERGY ENTERING AND PASSING ALONG

### C4.2.1 Ecosystems are open systems

p. 579

Energy and matter can both enter and leave an ecosystem. In a closed system, only energy can do so.

### C4.2.2 Sunlight sustains most ecosystems

pp. 579–580

Exceptions include caves, and ocean depths below the reach of light. There, organic matter brought in, or chemical energy used by bacteria, supplies the energy.

### C4.2.3 Chemical energy flows through food chains

pp. 580–581

Chemical energy passes to a consumer as it feeds on the previous organism in the chain.

### C4.2.4 Food chains and food webs

pp. 581–582

Chains and webs show feeding relationships. **Arrows** show the direction of transfer of energy and biomass.

### C4.2.5 Energy for decomposers

pp. 582–583

Decomposers receive energy as carbon compounds in faeces, dead parts of organisms and dead whole organisms.

## HOW ORGANISMS OBTAIN ENERGY AND CARBON

### C4.2.6 Autotrophs

p. 584

Autotrophs use an external energy source to make carbon compounds from simple inorganic molecules. Energy is needed for carbon fixation, and for reactions that build macromolecules.

### C4.2.7 Photoautotrophs and chemoautotrophs

p. 585

**Photoautotrophs** use light. **Chemoautotrophs** use oxidation reactions, which release energy. Iron-oxidizing bacteria are an example.

### C4.2.8 Heterotrophs

p. 585

Heterotrophs digest carbon compounds from other organisms, internally or externally, and assimilate the products to build the compounds they need.

### C4.2.9 Energy release in cell respiration

p. 585

Autotrophs and heterotrophs both release energy by oxidizing carbon compounds in cell respiration.

### C4.2.10 Trophic levels

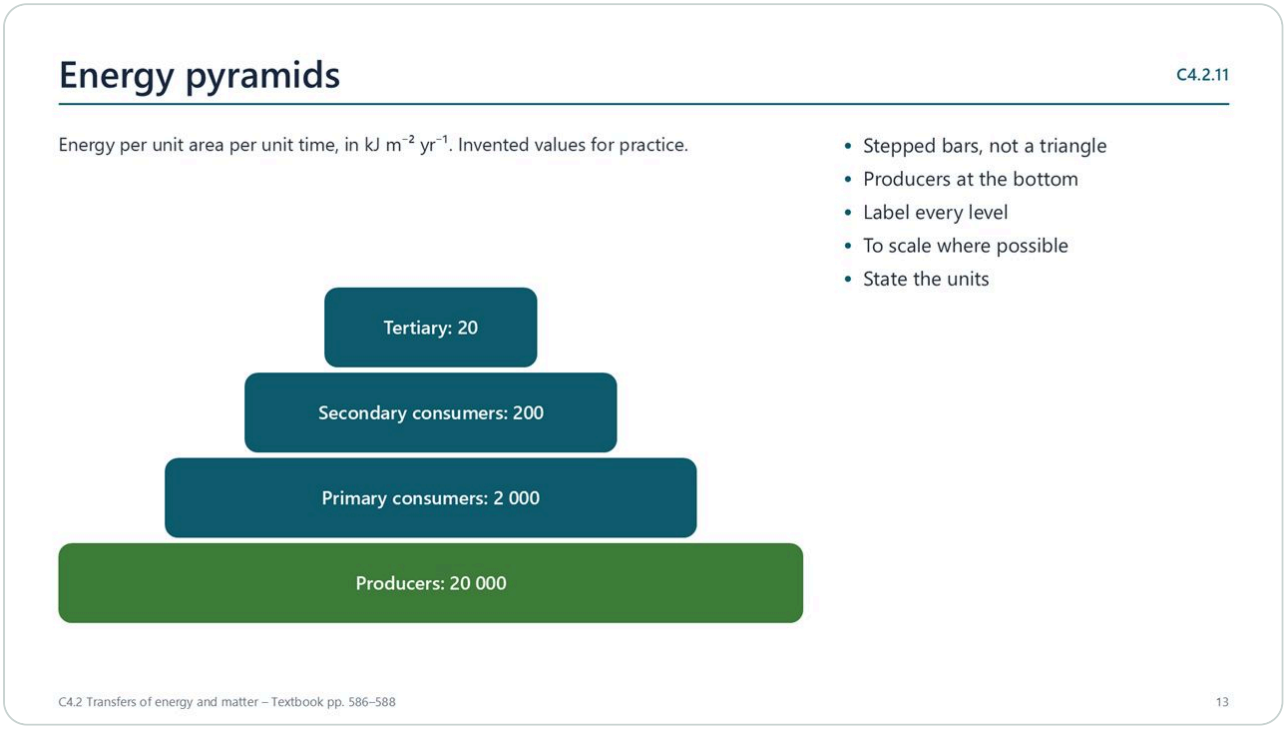
p. 586

Levels are **producer**, **primary consumer**, **secondary consumer** and **tertiary consumer**. Many organisms have a varied diet, so they can feed at more than one level.

## ENERGY LOST AT EACH STAGE

### C4.2.11 Energy pyramids

pp. 586–588




Pyramids are stepped and show energy per unit area per unit time, such as kJ $m^{-2}$ $yr^{-1}$. Producers are at the bottom.

### C4.2.12 Why energy falls along a chain

pp. 589–590

Some organisms or parts are not eaten, some food is not digested, and carbon compounds are oxidized in respiration. Decomposers and detritus feeders are not usually placed in food chains, though they do transform energy.

### C4.2.13 Heat loss

pp. 590–591

Energy transfers are not 100% efficient. Heat is produced when ATP is made in respiration, and when ATP is used in cells. This heat passes to the environment, and organisms cannot reuse it.

### C4.2.14 Limits on trophic levels

p. 591

At each stage there are fewer or smaller organisms, so there is less biomass. Energy per unit mass is not reduced. Too little total energy remains to support many levels.

### C4.2.15 Primary production

pp. 592–593

This is the build-up of carbon compounds in autotrophs, measured in g $m^{-2}$ $yr^{-1}$. Biomes vary in how much they accumulate.

### C4.2.16 Secondary production

pp. 593–594

This is the build-up of carbon compounds in heterotrophs. It is lower than primary production, because biomass is lost as carbon dioxide and water in respiration.

## THE CARBON CYCLE

### C4.2.17 Carbon cycle diagrams

pp. 594–595

Show how carbon is recycled by **photosynthesis**, **feeding** and **respiration**.

### C4.2.18 Sinks and sources

pp. 595–596

If photosynthesis exceeds respiration, an ecosystem has a net uptake and acts as a **sink**. If respiration exceeds photosynthesis, it has a net release and acts as a **source**.

### C4.2.19 Combustion

pp. 597–598

Burning biomass, peat, coal, oil and natural gas releases carbon dioxide. These stores formed at different times. Lightning can start fires, but human activity greatly raises the amount of combustion.

### C4.2.20 The Keeling curve

pp. 599–600

Carbon dioxide at Mauna Loa has been measured since 1958. The **long-term rise** reflects combustion. The **yearly rhythm** reflects photosynthesis, which is greater in the northern summer, and respiration, which exceeds it in winter.

### C4.2.21 Photosynthesis and respiration depend on each other

p. 601

Aerobic respiration uses oxygen from photosynthesis. Photosynthesis uses carbon dioxide from respiration. The amounts each year are huge.

### C4.2.22 All elements are recycled

p. 601

All elements that organisms use are recycled, not just carbon, and decomposers play a key role.

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